HAMZA EL MADANI

Junior AI / LLM Engineer | RAG · LLMOps · Python · ROS 2

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PROFILE

Computer Science B.Sc. candidate at Brandenburg University of Applied Sciences (thesis submitted; colloquium planned for early December 2026). Hands-on project experience building and evaluating local LLM applications, retrieval-augmented generation systems, Python backends, and ROS 2 robotics. More than six years of prior experience in electronics and technical maintenance. Open to full-time opportunities across Germany.

SELECTED PROJECTS & TECHNICAL RESULTS

THB StudentAssist — Multilingual RAG Chatbot | GitHub: THB-StudentAssist

Python · LangChain · Chroma · FastAPI · Ollama · LiteLLM · Docker · pytest

- Built hybrid retrieval combining vector search, a custom BM25 implementation, and Reciprocal Rank Fusion; citation-grounded answers with abstention for unsupported queries.
- Project-specific evaluation across 4 models and 14 questions × 3 runs: gemma3:4b scored 0.91 vs. 0.86 for gemma3:27b, using 2.8 vs. 16.7 GB VRAM.
- Delivered German, English and Arabic responses, streaming FastAPI endpoints, rate limiting, and 19 CI tests.

Robot LLMOps — Self-Hosted Gateway, Monitoring & Benchmarking | GitHub: llmops_robot

Python · LiteLLM · Ollama · Docker Compose · Prometheus · Grafana

- Implemented an OpenAI-compatible gateway with model aliases, fallback routing, and API-key authentication; monitored GPU, VRAM, RAM and temperature.
- Benchmarked 33 German/English robot commands; llama3.1:8b achieved 88% tool-and-argument accuracy and 0.18 s median latency in the documented test run.

Bachelor's Thesis — LLM Agent for Robot Object Navigation | GitHub: llm-mcp-robot-navigation

Python · ROS 2 Humble · MCP · Ollama · SAM · TurtleBot3 · pytest

- Compared five LLM/MCP integration architectures sharing perception and safety components; implemented three independent emergency-stop layers.
- Evaluated a three-skill closed-loop architecture in 110 runs: 8/9 solvable scenarios succeeded in all 10 trials each; 20/20 unreachable-target trials correctly identified.

AI-Powered Autonomous Robot Navigation | GitHub: AI-Powered-Autonomous-Robot-Navigation

Python · ROS 2 · Ollama · VOSK · SAM · OpenCV

- Developed voice-controlled object search on a physical TurtleBot3 using offline speech recognition, LLM-based target extraction, segmentation, visual search and reactive alignment/approach.

TECHNICAL SKILLS

AI & LLMs: RAG, hybrid retrieval, BM25, embeddings, LangChain, Chroma, Ollama, LiteLLM, MCP, tool calling, LLM evaluation

Backend & Operations: Python, FastAPI, Docker / Docker Compose, pytest, GitHub Actions, Prometheus, Grafana, Linux, Git

Robotics & Other Languages: ROS 2 Humble, TurtleBot3, OpenCV, SAM, VOSK; Java, C, C++, SQL

EDUCATION

B.Sc. Computer Science — Brandenburg University of Applied Sciences (TH Brandenburg) | Sep 2022 – Dec 2026 (expected)

Bachelor's thesis submitted; colloquium expected in early December 2026. Thesis: "Integration of an LLM Agent into the Decision-Making Logic of Mobile Robots Using Object Navigation as an Example."

University Diploma of Technology (DUT), Maintenance / Electromechanics — Université Cadi Ayyad, Morocco | Sep 2012 – Jun 2014

PROFESSIONAL EXPERIENCE

Sales Agent (alongside studies) — Berlin City Tour, Berlin | Nov 2022 – Aug 2026

Multilingual customer consultation and sales for international visitors; independently handled customer complaints.

Electronics Technician — Shems for Lighting, Safi, Morocco | Feb 2017 – Jan 2022

Installed electrical systems, performed preventive and corrective maintenance, and conducted systematic fault diagnosis.

Maintenance Technician — Highprotech, Casablanca, Morocco | Dec 2015 – Jan 2017

Maintained industrial machinery and contributed to improvements in machine operations.

LANGUAGES

German: DSH-3 · English: B2 · French: Fluent · Arabic: Native